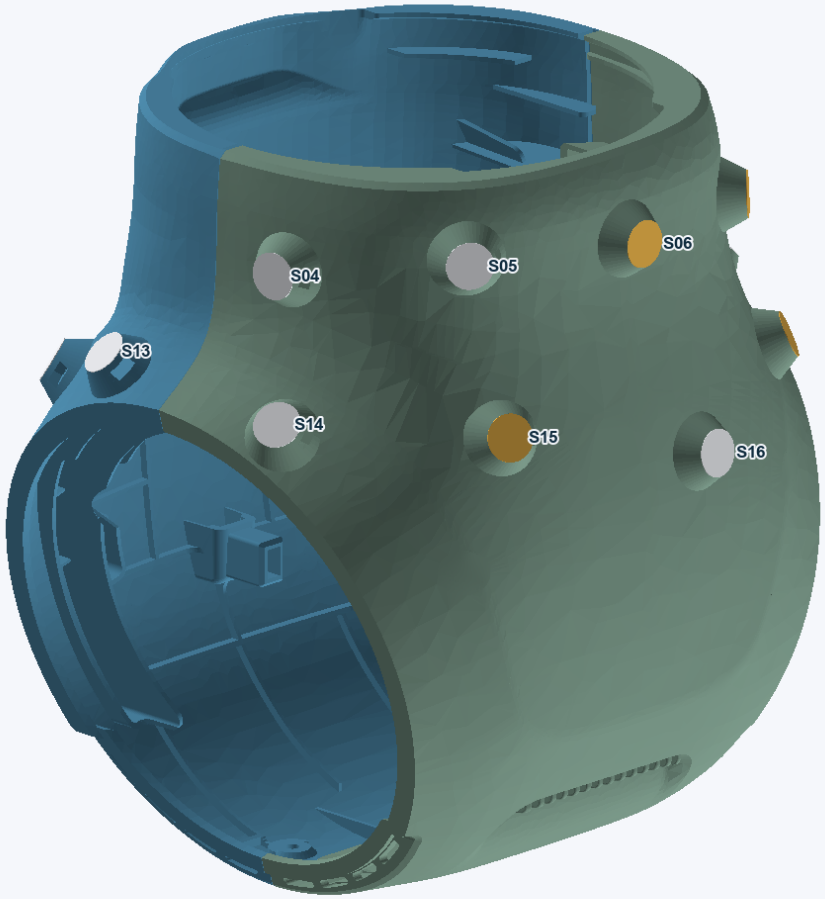
A3 dual-mode marker shell v3

17 stations / front seven unchanged / rear upper and middle rows: five each

A3 v3 / 17 stations / front 7 unchanged / rear 5 + 5

Old mounts removed | printable mesh geometry checked | physical validation pending



Gold: five restored rear stations. White: unchanged. All stickers are 12 mm x 0.20 mm.

Rear-only change: S06, S07, S08, S15 and S17 are restored at their original CAD locations. Rear upper row is S04-S08; middle row is S14-S18. Gold highlights the five restored stations. The front STL and its seven marker positions are unchanged.

Same v3 fit and hardware: flat 12 mm tops, no raised rims or recessed sticker pockets. A flush pin supports each 12 mm sticker over its central blind bore. Film plus adhesive is 0.20 mm. Ball mode retains the provisional side-loaded M3 captive-nut mount.

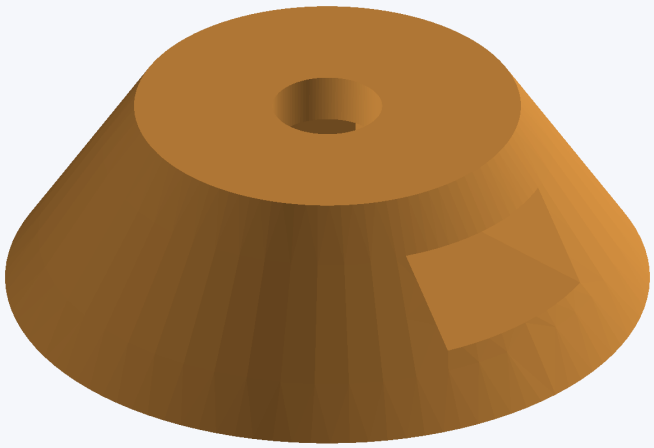
Release scope: checked STL geometry and nominal CAD transforms, not physical qualification. Printed fit, hardware retention, impact survival and camera visibility need testing. Five restored points have no observations in the supplied Motive export.

One station, two marker modes

No new hole through the pelvis-facing shell / ball hardware remains provisional

Flat 12 mm top / central screw bore preserved

No raised rim or sticker pocket | 3.4 mm bore | M3 hardware provisional | pin removed



Local hardware-fit illustration. Ball thread, nut and plug fit must be validated before shell release.

Station: flat 12.0 mm outside-diameter top; tapered 20.0 mm support foot below it. Central bore: 3.4 mm diameter, 4.6 mm blind axial depth. No raised guard or full-face sticker recess.

Side-loaded nut: provisional M3 nut, 5.5 mm across flats and approximately 2.4 mm thick. Pocket: 5.8 mm across flats by 2.65 mm high; top 1.5 mm below the seat. Insert from the outside side slot, not from the pelvis-facing interior. Verify actual nut fit and retention.

Sticker mode: print up to 17 pins, each 3.45 mm diameter by 1.30 mm long. Trial-fit fully flush, then centre a 12 mm sticker of 0.20 mm total thickness. Do not force the interference fit or sand the optical support plane.

Ball mode: remove sticker and pin; fit the verified ball stud and captive nut. Candidate stud penetration must not exceed 4.0 mm below the seat without new checks. Confirm thread engagement and seating. Do not drill through the blind floor or force an overlong stud.

Different optical centres: ball diameter does not determine its mounting thread or installed stand-off. Measure the ball/stud/spacer assembly before creating a ball-centre table and asset. The sticker table and old 6 mm registration stand-off are not ball-mode calibration.

Marker translations to pelvis_link

ROS: X forward / Y left / Z up; metres; CAD nominal, not physical accuracy

Sticker centres include 0.20 mm film above the top plane. Seat coordinates locate the screw axis at that plane, **not the ball sphere centre**. Original IDs and previous coordinates are unchanged. Display rounding is 0.000001 m, not an accuracy claim.

ID	Sticker X	Sticker Y	Sticker Z	Seat X	Seat Y	Seat Z
S04	-0.023681	0.072053	-0.061631	-0.023648	0.071856	-0.061636
S05	-0.067166	0.047266	-0.054582	-0.067006	0.047146	-0.054589
S06	-0.079001	-0.001320	-0.057604	-0.078802	-0.001319	-0.057610
S07	-0.063806	-0.052205	-0.060498	-0.063659	-0.052070	-0.060507
S08	-0.019875	-0.072036	-0.054669	-0.019844	-0.071838	-0.054673
S11	0.068681	0.013753	-0.093372	0.068482	0.013736	-0.093385
S12	0.055718	0.068988	-0.102706	0.055579	0.068885	-0.102807
S13	0.024198	0.082666	-0.090054	0.024146	0.082537	-0.090198
S14	-0.035526	0.082456	-0.094529	-0.035453	0.082323	-0.094659
S15	-0.077885	0.044045	-0.097023	-0.077708	0.043973	-0.097082
S16	-0.086616	-0.018974	-0.113904	-0.086419	-0.018963	-0.113934
S17	-0.065030	-0.065131	-0.099352	-0.064902	-0.065000	-0.099433
S18	-0.022710	-0.083580	-0.089898	-0.022665	-0.083449	-0.090042
S19	0.034664	-0.079647	-0.092672	0.034583	-0.079518	-0.092801
S20	0.066488	-0.030622	-0.093038	0.066293	-0.030580	-0.093060
S21	0.079132	0.034881	-0.156415	0.078934	0.034855	-0.156395
S22	0.079113	-0.034871	-0.156403	0.078915	-0.034846	-0.156383

Full transforms: the two CSV tables and XLSX include translations, qx/qy/qz/qw and outward normals for all 17 stations. Each transform maps marker-local points into pelvis_link. Local +z points outward; round-marker roll is a CAD convention.

Front unchanged: S11/S12/S13/S19/S20/S21/S22. **Rear:** S04-S08 and S14-S18. S01/S02/S03/S09/S10/S23/S24 stay absent. No marker is renumbered.

CAD registration and verification

The same nominal pelvis datum; no centroid reset or new axis correction

Datum provenance: the original STEP has no named robot root. Ten nominal original marker coordinates define the pelvis reference. A proper rotation, translation and shared old marker stand-off were jointly fitted to ten socket annuli (seven parameters, 30 coordinates). The fitted 6 mm stand-off applies only to those old registration markers.

Unchanged coordinate mapping:
pelvis_x_m = CAD_Z / 1000
pelvis_y_m = CAD_X / 1000
pelvis_z_m = (CAD_Y - 145.800095037096) / 1000
The CAD source is in millimetres; this is a right-handed rotation (determinant +1). The origin is the existing nominal pelvis_link datum, not the marker centroid.

Historical construction checks: maximum datum residual 4.122e-13 mm; held-out maximum 4.673e-13 mm; Jacobian rank 7/7. The archived 24-station STEP seat check had maximum discrepancy 9.995e-12 mm. These are numerical agreement checks, not physical micrometre accuracy.

Current table checks: all 17 formula-driven workbook rows agree with the archived CAD calculation within 1e-12. Increasing film thickness by 0.10 mm moves only sticker centres along the outward normals. The released input is restored to 0.20 mm. Previous 12 CSV rows are unchanged.

Current mesh checks: front 74,536 triangles; rear 99,132. Each has one closed, consistently wound component, no duplicate/degenerate triangles and no MeshLab-detected self-intersections. The assembled intersection volume is zero. All ten rear seats pass flat-top, 12 mm diameter, 3.4 mm bore and 4.6 mm blind-depth probes within 0.001 mm.

Surface retention: outside the five restored footprints, 84,425 rear surface samples are within 0.00000296 mm of the new mesh. The front file is byte-identical. This is a sampled retention check, not an absolute CAD tolerance. FreeCAD was not rerun on the new rear mesh.

Limits: consult CAD_registration.md, geometry_repair_status.md and printability_validation.json for exact hashes and historical repair limitations. The physical pelvis datum, installed fit and optical calibration remain unverified.

Printing and validation checklist

Use millimetres; verify the sliced print and installed hardware before operation

1 / Print the current mesh masters. Print printable/a3_marker_shell_v3_half_a.stl and half_b.stl separately. The model stays v3. Old invalid STEP conversions and partial previews are not manufacturing masters. The flush pin is unchanged; print up to 17 for sticker mode.

2 / Import at 100% scale in millimetres. Half footprints are approximately 176 x 194 mm before brim/support margins. In OpenSCAD use F5 for assembly; choose half_a or half_b for F6/export. With bed_oriented=true the open side faces the bed at Z=0. Print placement never changes the marker table.

3 / Inspect sliced layers. Select printer/material-specific supports and check thin ribs, bridges, the seam, structural screws, blind bores and nut slots. Keep support scars away from the flat optical faces. No printer-specific slicing or material strength qualification has run.

4 / Respect the local wall limit. Historical repair sampling found roughly 2.5 mm at most restored old-boss sites, but approximately 1.50 mm at a local front f1 edge gap. Ensure the slicer preserves it. This is not a certified global minimum thickness. Old f1 and b1-b5 bosses/bores remain removed.

5 / Fit on a secured robot. Check shell fit, structural fasteners, seam clearance, cable access, ventilation, nut retention and full joint travel. No through-hole/internal-nut redesign is included. Do not force an overlong ball stud or tight flush pin. Neither marker mode has an impact-survival rating.

6 / Check optical coverage. Test rear coverage, front/rear occlusion and grazing views with real cameras, arms and torso. The front seven stay unchanged; restored S06/S07/S08/S15/S17 are not confirmed visible by the existing source asset. No all-pose visibility guarantee is made.

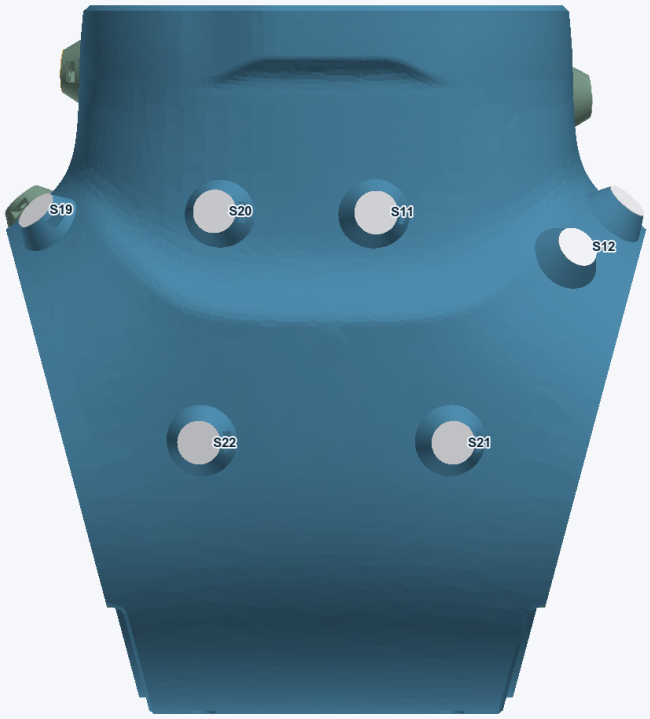
7 / Complete fresh calibration. Verify all 17 marker definitions and collect physical labeled-marker samples for each. They need not be simultaneously visible. Reject old 12-station receipts. Significant deviation from the expected identity correction is a discrepancy to investigate, not a value to force to zero.

Station identification: front and rear

Gold = restored rear points; white = unchanged points; illustration is not visibility evidence

Robot front / unchanged seven markers

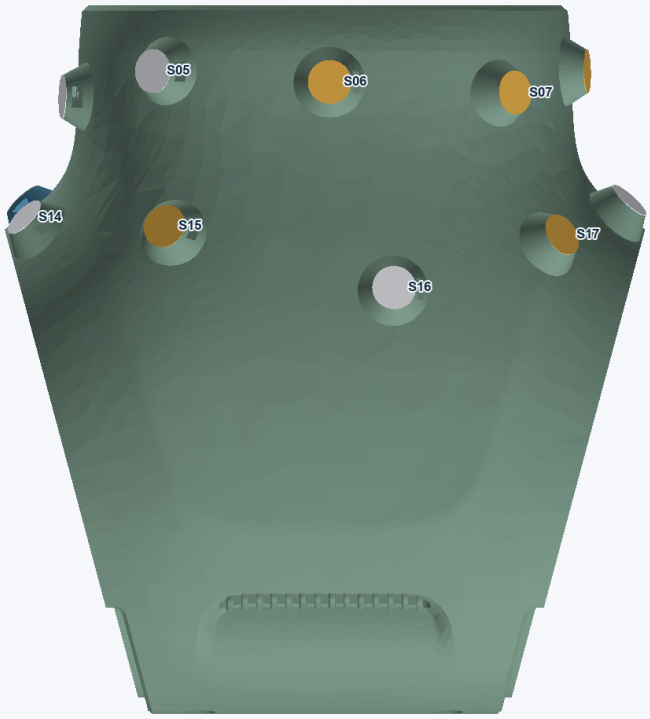
Old mounts removed | printable mesh geometry checked | physical validation pending



Gold: five restored rear stations. White: unchanged. All stickers are 12 mm x 0.20 mm.

Robot rear / S04-S08 and S14-S18

Old mounts removed | printable mesh geometry checked | physical validation pending



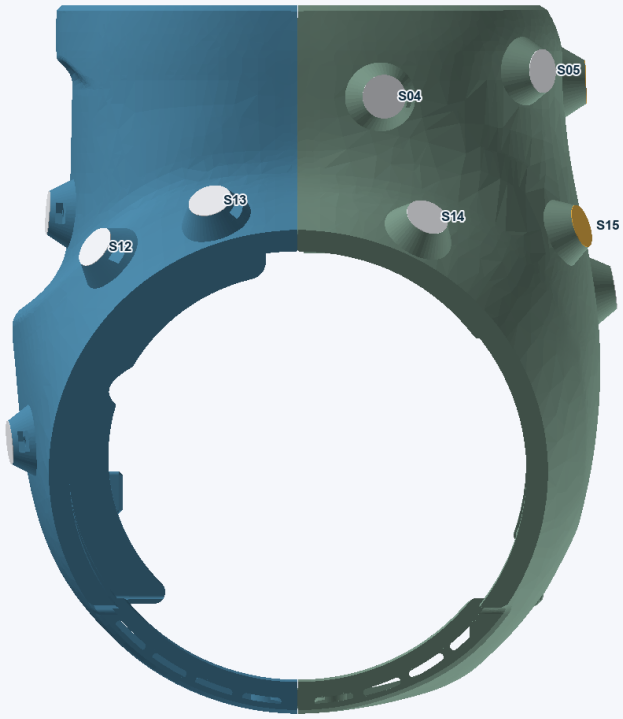
Gold: five restored rear stations. White: unchanged. All stickers are 12 mm x 0.20 mm.

Station identification: side arcs

Use the original Sxx labels; side stations complete each five-point rear row

Robot left / CAD +X

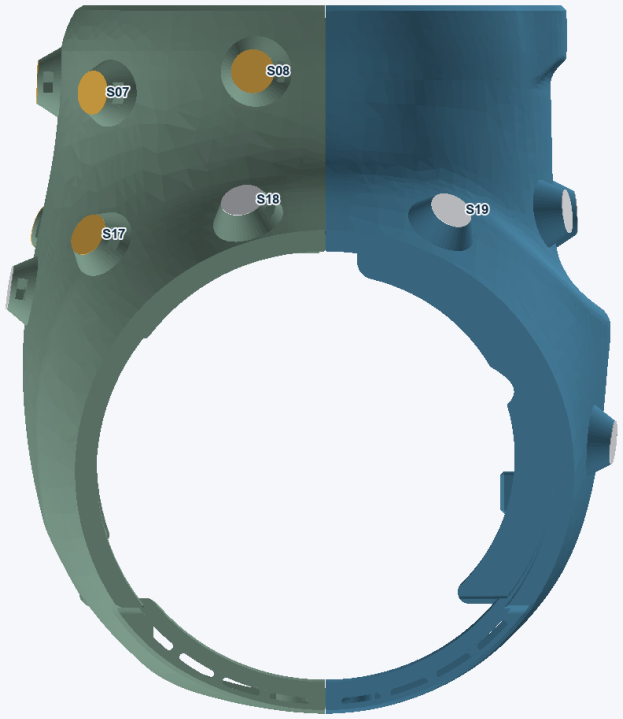
Old mounts removed | printable mesh geometry checked | physical validation pending



Gold: five restored rear stations. White: unchanged. All stickers are 12 mm x 0.20 mm.

Robot right / CAD -X

Old mounts removed | printable mesh geometry checked | physical validation pending



Gold: five restored rear stations. White: unchanged. All stickers are 12 mm x 0.20 mm.

OptiTrack asset and runtime frame

A3_v3_17_stickers.motive / P1 / streaming ID 9 / import and live validation pending

Mixed-provenance candidate: the original 12 measured offsets, diameters and member IDs are preserved in the aligned pelvis frame. Five new rear points are nominal CAD additions. Source exports remain untouched. The mapping below is for this candidate, not the separate ideal-CAD authoring order.

Station	Member	Source point	Evidence
S04	8	10	Measured; 0.858511 mm fit residual
S05	11	14	Measured; 1.107767 mm fit residual
S06	13	-	CAD nominal; not observed
S07	14	-	CAD nominal; not observed
S08	15	-	CAD nominal; not observed
S11	3	4	Measured; 1.330930 mm fit residual
S12	4	5	Measured; 1.224260 mm fit residual
S13	12	16	Measured; 1.646311 mm fit residual
S14	10	12	Measured; 0.765856 mm fit residual
S15	16	-	CAD nominal; not observed
S16	7	8	Measured; 1.191584 mm fit residual
S17	17	-	CAD nominal; not observed
S18	9	11	Measured; 1.556395 mm fit residual
S19	1	1	Measured; 1.595486 mm fit residual
S20	2	2	Measured; 1.802336 mm fit residual
S21	6	7	Measured; 1.853359 mm fit residual
S22	5	6	Measured; 1.324662 mm fit residual

Import gate: the learned constraint_graph cache is omitted, not fabricated. Native acceptance is unverified. Test-import in an isolated Motive project (source UUID is retained), confirm 17 markers and unique ID, then save a fresh native export. If rejected, author via Motive API from the supplied pivot-relative header/JSON; that path uses all-ideal CAD. See README.

Axes: native storage is [pelvis_x, pelvis_z, -pelvis_y]. Set Motive Streaming Up Axis = Z and keep MODELDEF Y-up-to-Z-up conversion enabled. Expected fixed P1-to-pelvis correction: xyz [0,0,0], xyzw [0,0,0,1]. Do not reset the pivot, recenter, apply the old offset or add a second 90-degree pose rotation.

Runtime: rename P1 to UCB_P1 or UCB_P2 for the selected main-branch table side; select stickers_v3. Restart NatNet and obtain a fresh live calibration/ucb_robot_to_pelvis.json receipt. The supplied identity JSON is unapproved reference data. Measured-subset RMS remains 1.395 mm, maximum 1.853 mm; nominal additions are not measurement evidence.